

EMPOWERING EDUCATION THROUGH AI-SUPPORTED E-LEARNING

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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Prof. Nuwan Kodagoda	15/03/2026	 2026/03/16

ABSTRACT (Maximum 300 words)

While multimedia content is increasingly prevalent in modern higher education, traditional e-learning platforms often treat audio and podcasts as passive mediums. This passive consumption frequently leads to student disengagement and suboptimal knowledge retention. Although recent advancements in AI-driven educational tools provide automated summaries, a significant research gap exists in facilitating true bidirectional interactivity; learners currently lack the ability to directly converse with and query the audio stream in real-time. To bridge this gap, this research proposes an interactive, AI-powered audio review engine designed to transform passive listening into a dynamic, active learning session. The proposed system utilizes a pipeline of specialized open-source models, integrating Automatic Speech Recognition (ASR) to accurately transcribe uploaded educational audio. Subsequently, an open-source Large Language Model (LLM) is employed to analyze transcripts, identify central themes, and generate structured summaries. A core feature of the solution is its interactive web interface, which allows users to navigate audio via transcript segments and ask context-specific questions, receiving immediate verbal explanations through an integrated Text-to-Speech (TTS) model. The expected outcome is a highly engaging, queryable audio document that significantly improves active comprehension by providing an immersive, conversational layer to standard audio files. Socially, this innovation advances inclusive, self-directed learning by catering to diverse auditory learning needs. Commercially, it disrupts the EdTech landscape by delivering a transparent, customizable, open-source alternative to existing proprietary, black-box AI educational tools.

- **Keywords:** Active Learning, Automatic Speech Recognition, Interactive Audio Review, Large Language Models, Educational Technology.

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LIST OF ABBREVIATIONS

- **AI** - Artificial Intelligence
- **API** - Application Programming Interface
- **ASR** - Automatic Speech Recognition
- **AWS** - Amazon Web Services
- **B2B** - Business-to-Business
- **B2C** - Business-to-Consumer
- **CDAP** - Comprehensive Design and Analysis Project
- **CoEAI** - Centre of Excellence for AI
- **DSR** - Design Science Research
- **EdTech** - Educational Technology
- **IEEE** - Institute of Electrical and Electronics Engineers
- **IRB** - Institutional Review Board
- **LKR** - Sri Lankan Rupee
- **LLM** - Large Language Model
- **LMS** - Learning Management Systems
- **MCQ** - Multiple Choice Question
- **NLP** - Natural Language Processing
- **SaaS** - Software as a Service
- **SDG** - Sustainable Development Goal
- **SE** - Software Engineering
- **TTS** - Text-to-Speech
- **UI** - User Interface
- **UX** - User Experience
- **WBS** - Work Breakdown Structure

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1. INTRODUCTION

1.1 Background and Context

In the current educational landscape, higher education institutions are heavily shifting towards digital and multimedia learning. However, they face a persistent challenge in maintaining student engagement and academic progression. The traditional "one-size-fits-all" pedagogical model is widely used, but it struggles to address the diverse learning needs, paces, and backgrounds of modern students. When students are provided with recorded lectures, educational podcasts, or long audio materials, the learning process often remains entirely passive.

This problem primarily affects university undergraduates and educators. Students coming from various non-traditional backgrounds or those with different learning styles are disproportionately impacted by these rigid learning structures. When learning is strictly passive, especially with audio or video content, students frequently experience disengagement and knowledge gaps. Global educational studies and e-learning statistics consistently show that student attention spans drop significantly after 10 to 15 minutes of passive listening, leading to higher dropout rates in online courses and a longer time-to-competency.

To address these issues, several technology-enhanced learning tools have been introduced to the market. Platforms like Google's NotebookLM and ChatGPT's Study Mode utilize large language models (LLMs) to process documents and generate tailored insights, summaries, and even audio overviews. However, these existing solutions have a major limitation: they treat audio output as a linear, one-way street. While tools like NotebookLM can generate an educational podcast from notes, they lack a bidirectional, conversational layer. If a student does not understand a specific point in the audio, they cannot interrupt, question, or interact with the audio stream in real-time. This leaves a massive gap in active comprehension.

To overcome these limitations, we propose the "AI-Powered Personalized Course Creation and Discovery Platform with Adaptive Learning Paths". This integrated platform replaces fragmented tools with a unified system where specialized AI modules operate around a central learner model. As a core part of this platform, my specific research focuses on developing an **Interactive, AI-powered audio review component**. Instead of passive listening, this component uses open-source Automatic Speech Recognition (ASR) to transcribe audio, an open-source LLM to analyze key themes, and an interactive web interface where users can click on any transcript segment to jump to that exact audio point. Furthermore, by integrating a Text-to-Speech (TTS) model, the system allows students to ask context-specific questions verbally and receive real-time audio answers directly from the learning material.

This project aligns directly with the United Nations Sustainable Development Goal 4 (Quality Education) by making learning more inclusive, equitable, and effective through personalized adaptation. It also promotes lifelong learning opportunities (SDG 4.4). Furthermore, this research falls under the Centre of Excellence for AI (CoEAI) research group and the Software Engineering (SE) specialization, as it requires core research in machine learning and natural language processing to build these adaptive, conversational interfaces.

1.2 Literature Review

Interactive audio processing—combining Automatic Speech Recognition (ASR), Large Language Models (LLMs), and Text-to-Speech (TTS)—has emerged as a transformative approach for educational technology. Unlike passive audio content (e.g., lectures, podcasts), interactive audio systems enable real-time querying, transcription, and content adaptation, fostering active learning. This review examines recent advances relevant to developing an AI-powered audio review component that competes with tools like Google NotebookLM while addressing its limitations in interactivity and language support.

Recent research has focused on integrating LLMs directly with speech modalities. Fathullah et al. [1] demonstrated that LLMs could be prompted with audio inputs to perform speech recognition without traditional ASR pipelines, enabling more flexible audio understanding. Tang et al. [2] extended this by developing models capable of simultaneous speech recognition and audio captioning, which is critical for generating summaries from lecture recordings. Lakomkin et al. [3] further improved end-to-end ASR by contextualizing predictions with pre-trained language models, enhancing accuracy in domain-specific educational content.

For real-time applications, Jia et al. [5] proposed an efficient streaming architecture that enables LLMs to process speech without end-of-utterance delays—essential for live classroom or self-study interactions. Nagpal et al. [4] explored reinforcement learning to adapt ASR for disordered speech, highlighting the potential for inclusive design. However, these systems primarily support English and major world languages, leaving a gap for code-switched queries in languages like Sinhala and Tamil mixed with English ("Sinhlish"/"Thanglish"), which are common in Sri Lankan higher education.

Multimodal frameworks are also advancing. Ma et al. [9] introduced SLAM-LLM, an open-source modular framework that unifies speech, language, and audio processing. This provides a viable foundation for building transparent, customizable educational tools—unlike proprietary systems such as NotebookLM. Cappellazzo et al. [7] showed that LLMs can be strong audio-visual speech recognizers, suggesting potential for future video-integrated extensions. Li et al. [8] demonstrated LLM-based emotion recognition with ASR error correction, which could be adapted to detect learner confusion or engagement from voice tone.

Despite these advances, existing systems—including NotebookLM—remain largely **linear and non-interactive**. They generate podcasts or summaries but do not allow users to **query the audio in real time, jump to concepts via transcripts, or receive spoken, context-aware answers**. Furthermore, no major platform supports **code-switched South Asian languages** natively in audio-based learning.

This project addresses these gaps by building an interactive audio review component that:

- Uses **open-source ASR + LLM + TTS models** (inspired by SLAM-LLM [9]) for full transparency.

- Enables **real-time, queryable audio transcripts** (building on streaming ASR [5]).
- Supports **Sinhala/Tamil-English code-switched speech**—a first in this domain.
- Provides **voice-based Q&A and assessment generation**, transforming passive listening into active learning.

By grounding the design in recent IEEE research and targeting specific gaps in interactivity and language support, this component directly contributes to the project’s goal of surpassing NotebookLM’s capabilities.

1.3 Research Gap

Current audio-based learning tools, including platforms like Google NotebookLM, remain fundamentally **linear and passive**. They function as closed, proprietary systems that deliver content but **do not allow learners to dynamically query the audio in real-time** or jump to specific concepts. A **bidirectional, conversational layer** must be integrated directly into the audio stream to transform passive listening into an active learning and analysis session.

Another critical limitation is the **lack of support for code-switched languages**. Most commercial platforms are optimized for standard English and fail to accommodate diverse student populations. Students in Sri Lanka frequently communicate using mixed languages—**Sinhala-English (Sinhlish)** and **Tamil-English (Thanglish)**—yet current educational systems lack a dedicated natural language processing (NLP) engine to handle such queries. Addressing this gap will make learning platforms significantly more accessible to non-traditional learners.

Furthermore, existing approaches function as **single-purpose, fragmented tools** that separate content delivery from assessment. They lack a **transparent, customizable architecture** that integrates Automatic Speech Recognition (ASR), Text-to-Speech (TTS), and Large Language Models (LLMs) into a cohesive, open-source ecosystem. Without this integration, systems cannot provide **real-time, voice-based Q&A directly from the audio stream**.

Novelty of This Project

This research introduces three key innovations:

Gap Addressed	Novel Solution
Linear, passive audio	Fully queryable audio document with real-time voice Q&A
No local language support	First Sinhala/Tamil-English code-switched NLP engine
Fragmented, proprietary tools	Open-source pipeline integrating ASR + LLM + TTS (inspired by SLAM-LLM)

By shifting from linear audio to interactive queryable media and supporting code-switched languages, this project develops a **diverse, engaging, and personalized AI-powered audio review model** that surpasses current proprietary tools.

2. OBJECTIVE

2.1 Main Objective

To engineer a modular and interactive AI-powered study suite using open-source models that measurably outperforms current linear learning platforms in the key areas of audio engagement, visual knowledge exploration, and personalized assessment.

2.2 Specific Objectives

To achieve the main objective, the research is divided into the following measurable specific objectives, aligned with the individual components of the proposed system:

1. To develop an intelligent, dynamic infographic generator that creates interactive and data-rich visual narratives from structured notes by utilizing open-source Large Language Models (LLMs) to extract key entities and relationships.
2. To engineer an interactive, AI-powered audio review component that transforms passive multimedia listening into an active analysis session by integrating Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) models to enable real-time, bidirectional voice-based Q&A.
3. To build an Adaptive Quiz and Assessment Generation Engine that utilizes a knowledge-tracing model to personalize multi-format questions based on real-time learner performance and identified knowledge gaps.
4. To develop a Multilingual Handwritten Notes Extractor and Knowledge Base Organizer capable of extracting text from handwritten notes in English, Tamil, and Sinhala, and converting them into structured, interactive study materials.
5. To evaluate the integrated system's technical and educational efficacy by measuring transcription accuracy, response latency, pre- and post-test scores, and user engagement metrics against existing proprietary e-learning tools.

3. METHODOLOGY

3.1 Research Approach

To craft a practical solution to the real-world problem of passive e-learning, this project adopts a Design Science Research (DSR) approach. This approach is highly suitable as it involves the systematic creation and evaluation of an IT artifact—in this case, the integrated AI-powered learning platform. By iteratively designing, developing, and testing the system, we can ensure the final product directly addresses the identified gaps in current educational tools.

3.2 Technical Methods and Algorithms

For the specific development of the Interactive Audio Review component, the system utilizes a pipeline of specialized natural language and speech processing algorithms:

- **Transcription Phase:** The system takes uploaded educational audio or podcasts and uses an open-source Automatic Speech Recognition (ASR) model to transcribe the speech into text.
- **Analysis Phase:** A natural language processing pipeline powered by an open-source Large Language Model (LLM) processes the transcript to extract key concepts, identify central themes, and generate structured summaries.
- **Interactive Q&A Phase:** A custom algorithm links the transcript text directly to audio timestamps. When a user queries the system, the LLM generates a contextual answer, which is then converted into voice using an integrated open-source Text-to-Speech (TTS) model.

3.3 Tools and Platforms with Justification

The following technologies will be utilized to build the system:

- **Python:** Selected as the primary programming language due to its robust ecosystem for AI and machine learning.
- **Hugging Face Transformers & PyTorch:** These libraries will be used to implement and fine-tune the open-source ASR, LLM, and TTS models. They provide the flexibility needed compared to closed, proprietary APIs.
- **React.js & Node.js:** React will be used to build the interactive web interface where users can click on transcript segments to jump through the audio stream. Node.js will handle the backend API routing.

- MySQL / MongoDB & AWS: A scalable database (like MySQL or MongoDB) will be used to store learner profiles and transcripts. Cloud platforms like AWS will be used to deploy the heavy AI models to ensure minimal latency.

3.4 Data Collection and Ethical Considerations

Given the privacy constraints regarding real student educational records, a triangulated hybrid data collection strategy will be used:

- Synthetic Data Generation: We will utilize Large Language Models (like Gemini 3 Pro or Deepseek) to generate simulated structured course content and student chat logs, including localized Thanglish/Sinhlish queries, to train the conversational models.
- Controlled Pilot Study: A closed-group pilot study will be conducted with volunteer undergraduates to gather human-in-the-loop feedback on UI usability and audio interaction accuracy.
- Ethical Considerations: The development phase poses minimal ethical risk as it uses synthetic and public datasets, though checks for algorithmic bias will be documented. For the pilot study, we will obtain ethical clearance from the Institutional Review Board (IRB). Strict protocols will be followed, including informed consent, data minimization (collecting only necessary interaction logs), and ensuring the right to withdraw at any time.

3.5 Validation Strategy and Performance Metrics

The audio review component will be evaluated using both technical and user-centric performance metrics:

- Technical Metrics: The system's backend performance will be validated by measuring transcription accuracy (e.g., Word Error Rate) of the ASR model and the response latency of the voice Q&A features.
- User Metrics: The educational efficacy will be evaluated by measuring summary relevance and user engagement metrics collected during the pilot study

4. HIGH-LEVEL SYSTEM ARCHITECTURE

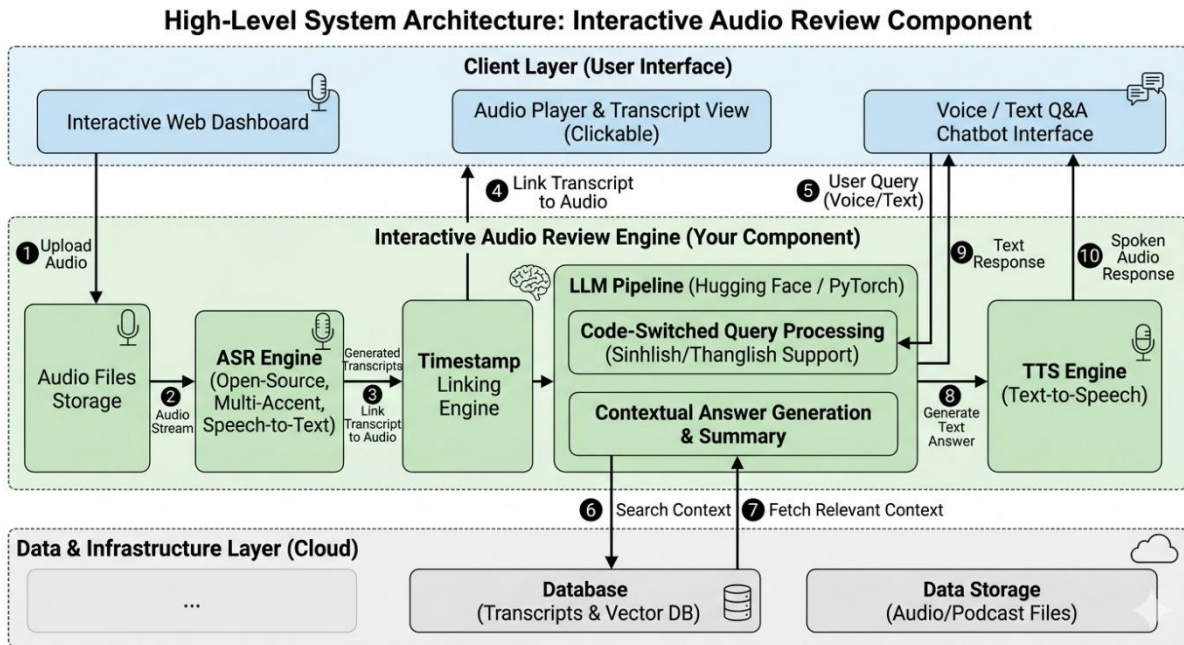


Figure 1: High Level System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders for this system are:

- **University Students (Undergraduates):** The primary users who will interact with the system to consume educational content, specifically those requiring active engagement to overcome passive learning habits.
- **Educators/Lecturers:** Users who will upload course materials, audio recordings, and podcasts to be processed and made interactive by the system.

5.2 Functional Requirements

To achieve the goals of the AI-powered audio review component, the system must fulfill the following functional requirements:

1. The system must allow users to upload educational audio files or podcasts and automatically transcribe them using an open-source Automatic Speech Recognition (ASR) model.
2. The system must generate and display an interactive text transcript alongside the audio player.
3. The system must allow users to click on any specific segment of the transcript to instantly jump to that exact timestamp in the audio stream.
4. The system must allow users to ask context-specific questions based on the transcript segment, supporting queries in English, Sinhlish, and Thanglish.
5. The system must analyze the audio transcripts to automatically generate structured summaries and identify central themes for the learner.
6. The system must be able to verbally read out summaries and answer user queries using an integrated Text-to-Speech (TTS) model.

5.3 Non-Functional Requirements

1. **Performance (Speed & Latency):** The system must ensure minimal response latency during the bidirectional voice-based Q&A process. Audio transcription and AI-generated verbal responses must be processed quickly to maintain a real-time conversational flow.
2. **Usability:** The interactive web interface must be highly intuitive, allowing students to seamlessly navigate between the audio player, transcript, and query chat without any distractions.

3. **Security and Data Privacy:** The system must ensure that any interaction logs or implicit feedback collected for model performance tuning are strictly minimized and stored securely, without compromising sensitive personal student records.
4. **Reliability and Accuracy:** The ASR model must maintain a high level of transcription accuracy for educational content, and the LLM must provide contextually relevant and hallucination-free summaries

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Initially, the target market is focused on the local higher education sector in Sri Lanka, specifically universities and large-scale tuition institutes. The primary customer personas are:

- **B2C (Students):** Undergraduates who struggle to review lengthy recorded lectures and need an interactive, time-saving tool for exam preparation.
- **B2B (Institutions):** Educational institutes looking to upgrade their Learning Management Systems (LMS) with AI features to increase student engagement.

6.2 Value Proposition Currently, students spend hours passively listening to audio notes or recorded lectures. Our interactive audio review component transforms this passive experience into an active, two-way conversational learning session. Furthermore, unlike foreign proprietary tools, our system natively supports local mixed-language queries (Thanglish and Sinhlish), making it highly relatable and effective for the Sri Lankan student demographic.

6.3 Revenue Model

The product will follow a standard Software as a Service (SaaS) model with two streams:

- **Freemium Model for Students:** Basic audio transcription is free, but advanced features like bidirectional AI voice Q&A and unlimited audio hours will require a monthly subscription.
- **Institutional Subscriptions:** Selling API access or bulk licensing to private universities and tuition platforms.

6.4 Cost Estimation

Since this is an AI-heavy application, the primary operational costs will be cloud infrastructure (e.g., AWS or Azure) to host the Large Language Models (LLMs) and Speech-to-Text models. Based on initial estimations for a local pilot launch, the monthly server and maintenance cost is projected at approximately LKR 120,000. Over a standard 10-month academic cycle, the total estimated seed cost required to run the platform is LKR 1.2 Million.

6.5 Pricing Strategy

To quickly penetrate the local student market, we will use a competitive freemium pricing strategy. The premium tier will be priced affordably for university students (e.g., LKR 900 - 1,500 per month). Institutional pricing will be tiered based on student headcount and audio processing volume.

6.6 Competitive Advantage and Financial Projection When compared to **existing** global solutions like Google NotebookLM , our competitive advantage is the localized language support and the real-time, bidirectional voice interaction layer. By successfully capturing the local EdTech market and gradually expanding to the South Asian region, we project that the product's valuation can scale up to 30 Million within a 5-year timeline.

6.7 Intellectual Property Considerations To protect our commercial interests, we will follow standard software IP protection methods. While we use open-source base models, our specific algorithm that maps transcripts to audio timestamps, along with our localized fine-tuning datasets, will be kept as trade secrets. The user interface and branding will be trademarked.

7. BUDGET AND JUSTIFICATION

Developing the interactive, AI-powered audio review component is primarily a software-based endeavor; therefore, there are no physical hardware components associated with the core implementation. To ensure the successful development of the speech recognition and conversational layers, we plan to utilize cloud computing.

Below is the estimated budget required to complete this specific component of the research project, kept well within the feasible limits of an undergraduate study.

Expense Category	Description	Estimated Amount (LKR)
Hardware Costs	Personal laptop and mobile device for development and testing.	0.00
Software Licenses	Open-source LLMs, ASR, and TTS models.	0.00
Cloud Services & APIs	Cloud processing capacity for model training and inference.	12,000.00
Internet & Connectivity	High-speed data for downloading audio datasets and models.	6,000.00
Learning Materials	Access to specialized online courses and research papers.	4,000.00
Human Effort Cost	Academic research and development hours.	0.00
Total Estimated Budget		22,000.00

Table 1: Budget

7.1 Justification for Each Expense

- **Hardware Costs (0.00 LKR):** As this is a software-centric module, all core programming and initial localized testing will be conducted using my existing personal ASUS laptop. No dedicated physical servers or external hardware are required for this phase.
- **Software Licenses (0.00 LKR):** The core novelty of this component relies on its transparent, open-source architecture. By strictly utilizing open-source models for Automatic Speech Recognition (ASR), Text-to-Speech (TTS), and text analysis, we completely eliminate the need for expensive proprietary software licenses.

- **Cloud Services & APIs (12,000.00 LKR):** This is the primary source of cost for the project. Generating foundational synthetic data requires interacting with advanced models like Gemini 3 Pro via Knowledge Distillation. Furthermore, running the open-source LLM to analyze transcripts for key concepts and processing audio requires the capacity of cloud-based virtual machines or services like Google Colab Pro, as standard local hardware is insufficient for high-speed inference.
- **Internet & Connectivity (6,000.00 LKR):** Continuous, high-bandwidth internet access is required to interface with cloud environments, download heavy audio files and podcasts , and support the real-time voice-based Q&A feature testing.
- **Learning Materials (4,000.00 LKR):** To gain the essential specialized domain expertise, funds are allocated for enrolling in specialized online courses on platforms like Coursera or edX focusing on AI in Education and Natural Language Processing.
- **Human Effort Cost (0.00 LKR):** Because this system is being developed as a mandatory IT4010 final-year research project, the coding and testing hours contributed are strictly considered academic effort and are not financially billed.

8. WORK BREAKDOWN STRUCTURE (WBS)

To ensure the successful delivery of the "AI-Powered Personalized Course Creation and Discovery Platform," the project is modularized into four distinct, yet interconnected, AI components. The Work Breakdown Structure (WBS) is designed to allocate an end-to-end development pipeline to each researcher, ensuring equal technical complexity and a balanced workload.

8.1 Task Breakdown Per Member

- **Member 1: Kojithan P.Y (IT22264220) – Interactive Infographic Generator**
 - **Task 1.1:** Design a schema to extract and structure key entities, relationships, and statistics from user notes using an open-source Large Language Model (LLM).
 - **Task 1.2:** Implement multiple visualization templates (hierarchical, timeline, network) using open-source libraries.
 - **Task 1.3:** Develop a dynamic interface allowing users to filter data, click elements to drill down into source quotes, and toggle visualization types.
 - **Task 1.4:** Integrate a rule-based or lightweight AI model to suggest the most suitable chart type.
 - **Task 1.5:** Enable real-time collaborative features for simultaneous viewing and annotation.
 - **Task 1.6:** Create an export module for interactive HTML or static image publishing.
- **Member 2: Baskaran V (IT22172600) – Interactive AI-Powered Audio Review Component**
 - **Task 2.1:** Implement automatic speech recognition (ASR) to transcribe uploaded audio/podcasts.
 - **Task 2.2:** Develop an NLP pipeline using an open-source LLM to analyze transcripts for key concepts, structured summaries, and central themes.
 - **Task 2.3:** Build an interactive web interface with clickable transcript segments for context-specific Q&A.
 - **Task 2.4:** Integrate an open-source Text-to-Speech (TTS) model for verbal AI answers.
 - **Task 2.5:** Conduct system evaluation based on transcription accuracy, response latency, and user engagement.

- **Member 3: Nishara T (IT22223876) – Adaptive Quiz and Assessment Generation Engine**
 - **Task 3.1:** Develop a Knowledge Tracing Model to infer concept mastery from interaction history.
 - **Task 3.2:** Implement an NLP pipeline to auto-generate diverse question types (MCQ, fill-in, short answer).
 - **Task 3.3:** Create an algorithm to dynamically adjust quiz difficulty and focus based on the learner model.
 - **Task 3.4:** Integrate a feedback mechanism offering Socratic hints and scraped web study resources.
 - **Task 3.5:** Evaluate the engine's impact on learning efficacy via pre/post-test scores.

- **Member 4: Sarmitha S (IT22637482) – Multilingual Handwritten Notes Extractor**
 - **Task 4.1:** Collect handwritten notes across English, Tamil, and Sinhala.
 - **Task 4.2:** Preprocess images (denoise, adjust contrast) and prepare labeled training data.
 - **Task 4.3:** Train or fine-tune an AI model for multilingual handwriting recognition.
 - **Task 4.4:** Clean, structure, and convert extracted text into study-ready Canva notes.
 - **Task 4.5:** Store structured outputs in a searchable digital knowledge base.

8.2 Timeline Alignment The project spans the 2026 academic year and is divided into four synchronized phases to ensure all components integrate smoothly into the Central Knowledge Orchestrator:

1. **Phase 1: Planning & Data Collection (Months 1-2):** Literature review, UI/UX design, synthetic data generation, and gathering multilingual note datasets.
2. **Phase 2: Core Model Development (Months 3-5):** Independent development of LLM extraction schemas, knowledge tracing models, and handwriting recognition algorithms.
3. **Phase 3: Integration & UI Implementation (Months 6-7):** Connecting the four standalone AI modules to the unified Personalized Learning Interface dashboard.
4. **Phase 4: Testing & Pilot Study (Months 8-10):** Executing the controlled pilot study with undergraduate volunteers to evaluate performance, followed by ethical clearance reporting and final documentation.

8.3 Balanced Workload Distribution The workload is equitably distributed by assigning each member a complete, vertically integrated subsystem. Every researcher is responsible for their module's data ingestion (e.g., text, audio, images), underlying AI/ML processing (e.g., NLP extraction, speech recognition, computer vision), user interface integration, and individual performance evaluation. This ensures that all four members engage deeply with core AI domains and software engineering practices without creating technical bottlenecks or unequal dependencies

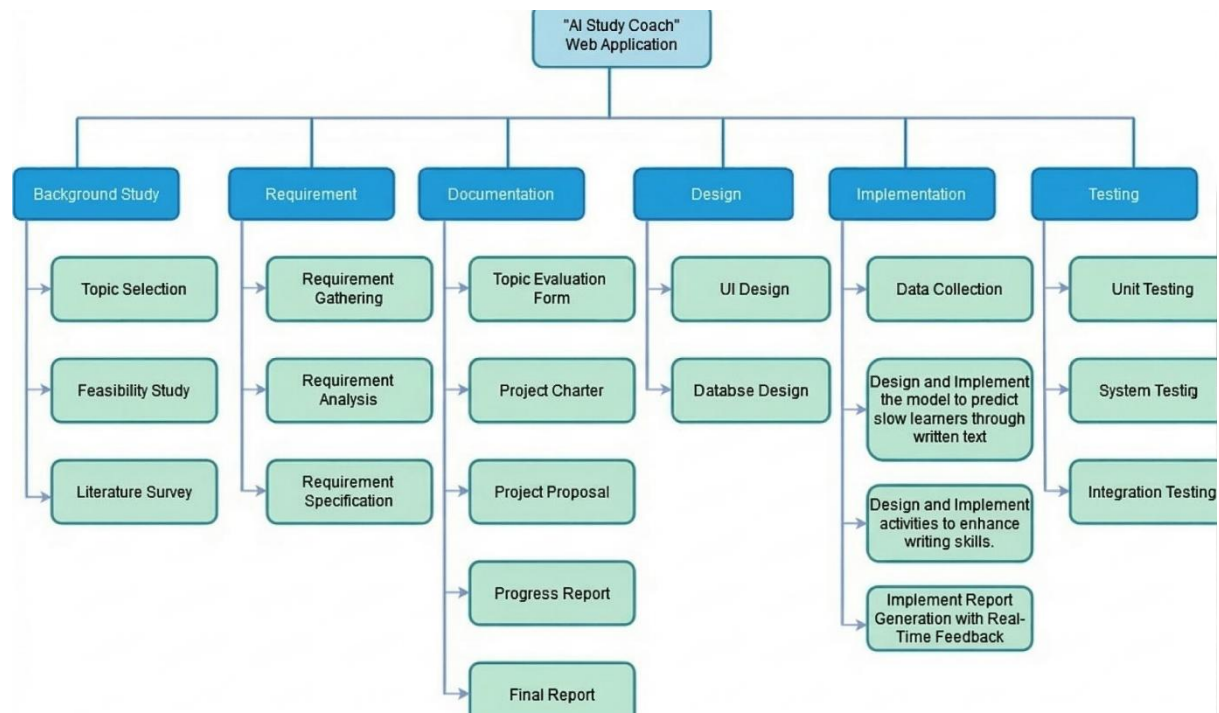


Figure 2: WBS

9. GANTT CHART

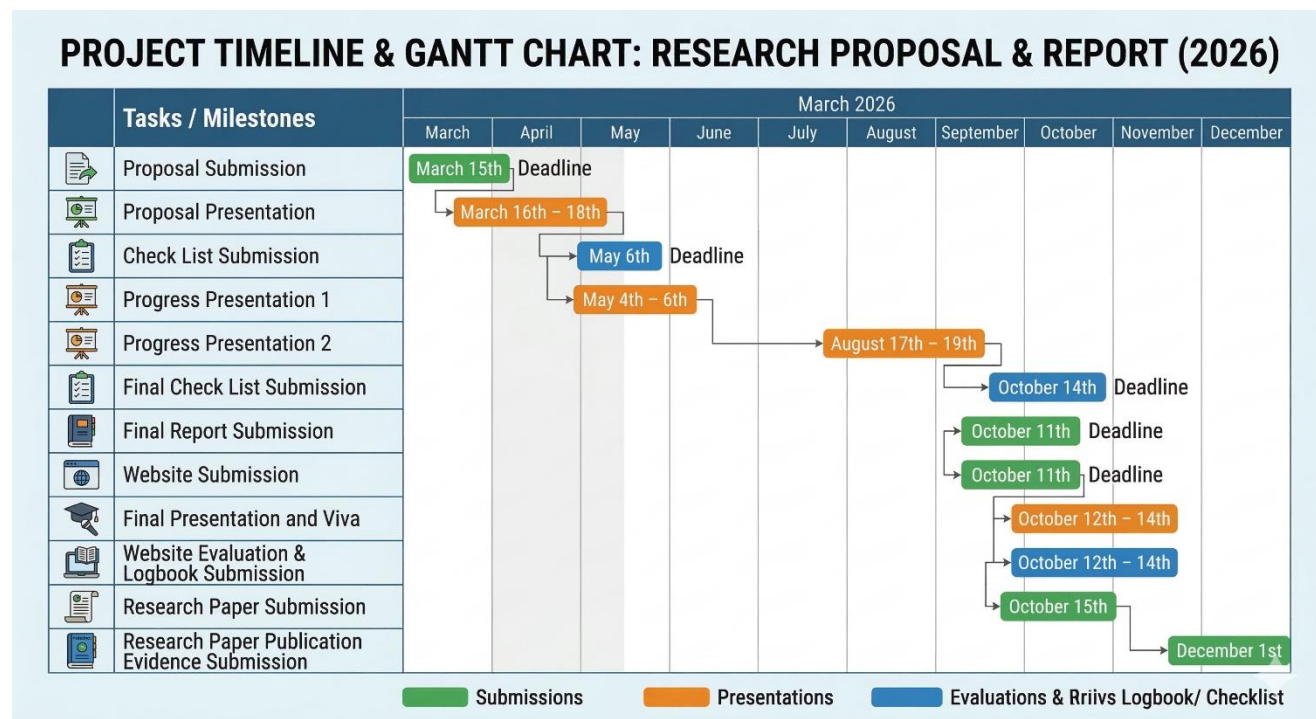


Figure 3: GANTT Chart

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11.APPENDICES

- **Survey instruments**

1. Online Questionnaires (via Google Forms)

What it is: Digital surveys given to students before and after using your interactive charts.

Why use it: To collect hard numbers and quantitative data (e.g., rating usability on a scale of 1to5).

Example: "Rate how easy it was to switch from the Timeline View to the Network View."

2. Structured Interviews

What it is: A one-on-one conversation using a strict, pre-written list of specific questions.

Why use it: To make sure you get qualitative answers on specific features you built (like the AI chart recommendation).

Example: "Did the AI correctly suggest the best chart type for your notes? Please explain why or why not."

3. Unstructured Interviews

What it is: An open, free-flowing conversation with the user, without a strict script.

Why use it: To discover unexpected insights. Sometimes users interact with a visual chart in ways you didn't plan for. This allows the user to freely share their feelings, frustrations, or "aha!" moments.

- **Example:** "Tell me about your experience exploring the data today. What stood out to you?" (And then you just listen and let the conversation flow naturally).

- Detailed Diagram

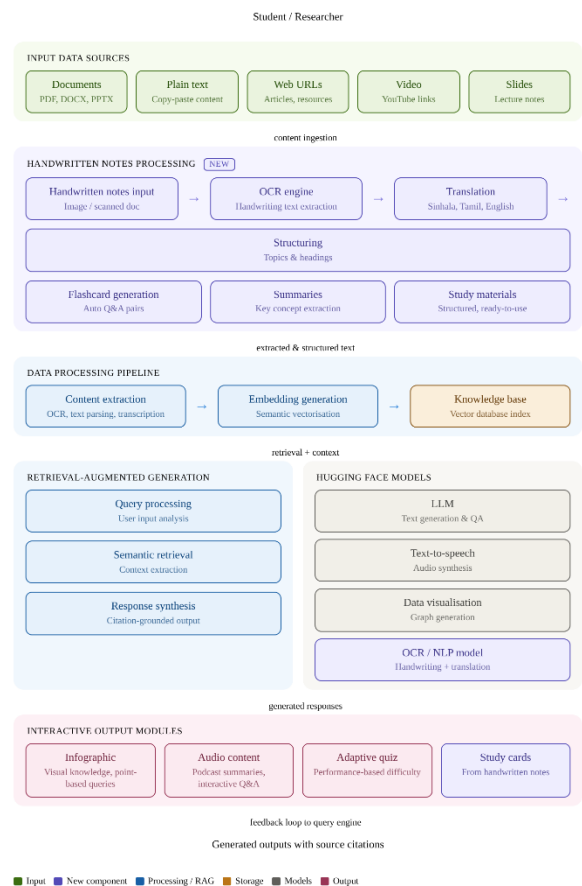


Figure 4: Detailed Diagram

- Preliminary UI mockups

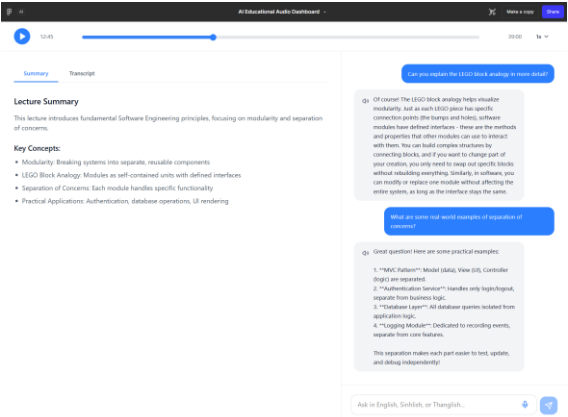


Figure 6: UI1

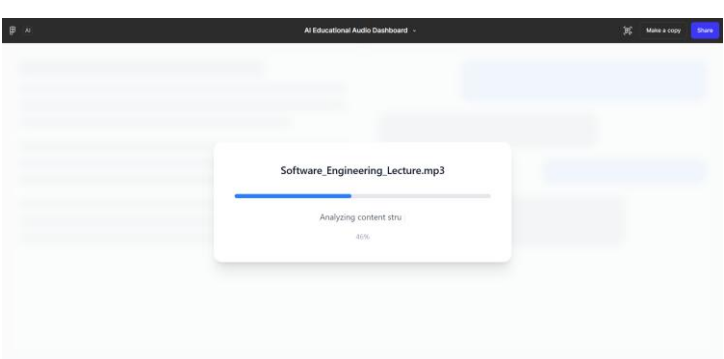


Figure 5: UI2

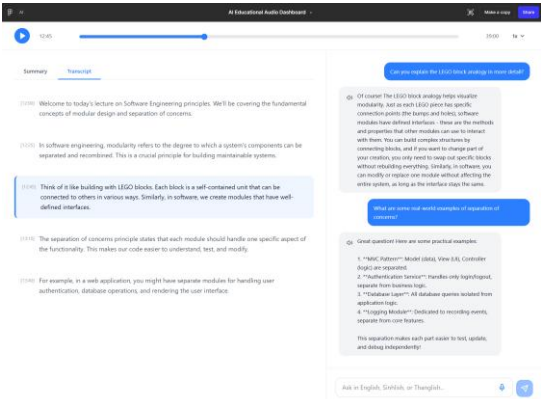


Figure 8: UI3

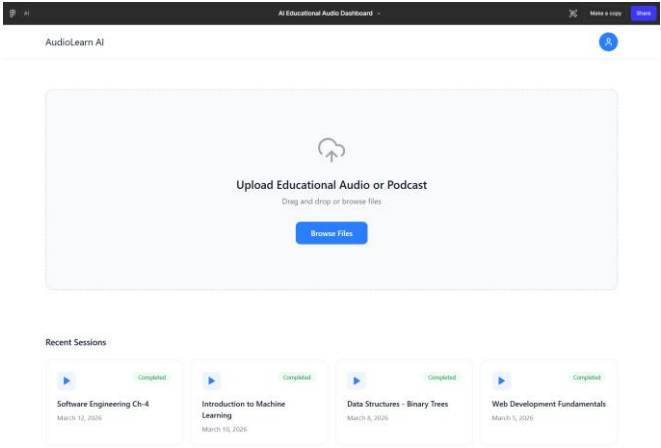


Figure 7: UI4

- **Risk Analysis**

1. Technical Risk: AI Transcription & Language Processing Errors

- **The Risk:** The open-source Automatic Speech Recognition (ASR) model might transcribe the educational audio inaccurately, especially with technical jargon or local accents. Furthermore, the LLM might struggle to accurately understand code-switched user queries (Sinhlish/Thanglish).
- **The Mitigation:** I will train the models using synthetic data generation (including simulated Sinhlish/Thanglish queries) to improve accuracy. Additionally, the user interface will allow students to visually verify the transcript, and I will implement robust prompt engineering to guide the LLM's understanding.

2. Performance Risk: High Latency in Voice Q&A

- **The Risk:** Generating real-time, bidirectional voice answers using the integrated Text-to-Speech (TTS) and LLM models might cause delays, disrupting the conversational flow and frustrating the learner.
- **The Mitigation:** I will deploy the heavy AI models on scalable cloud infrastructure (like AWS or Google Colab Pro) to ensure high-speed inference rather than relying on standard local hardware.

3. Integration Risk: Audio-Transcript Synchronization Failures

- **The Risk:** The custom algorithm responsible for linking the transcript text directly to the audio timestamps might fail or drift, causing the audio player to jump to the wrong segment when a user clicks on the text.
- **The Mitigation:** I will thoroughly test the timestamp linking engine across various audio formats and lengths during the controlled pilot study. The system will also provide fallback manual navigation controls on the interactive web dashboard.

4. Ethical Risk: Data Privacy and Audio Storage Concerns

- **The Risk:** Processing user-uploaded educational podcasts and logging voice queries could accidentally collect or expose sensitive institutional materials or personal student data.
- **The Mitigation:** I will implement strict data minimization protocols. The system will only collect necessary interaction logs (such as response latency and transcription accuracy metrics) and will securely handle data without compromising sensitive personal student records.